Duty at home

DM3 · A family of curves · 45 min.

For every real a, the function is considered f_a defined by $f_a(x) = \frac{ax^2+1}{x^2+1}$. We note $\mathcal{C}_a$ its curve representative.

1. Determine the definition set of f_a, then its limits in $-\infty$ and in $+\infty$.
2. Show that $f_a(x) - a = \frac{1-a}{x^2+1}$. Depending on the value of a, specify the $\mathcal{C}_a$ in relation to the right of equation $y = a$.
3. Studying variations in f_a distinguishing cases $a < 1$, $a = 1$ and $a > 1$. Clarification of possible extremema and where they are affected.
4. Take $a = 3$. Resolve, for $x \geq 0$, the inequitable $|f_3(x) - 3| \leq 0{,}02$.